



// PROPOSAL · PRELIMINARY RESULTS · MECHANISTIC INTERPRETABILITY

# Dictionary learning is all you need

Keira Wiechecki

WAKARU · KAW504@NYU.EDU

Jade Zaslavsky

WAKARU · JADEOFNAMELESS@GMAIL

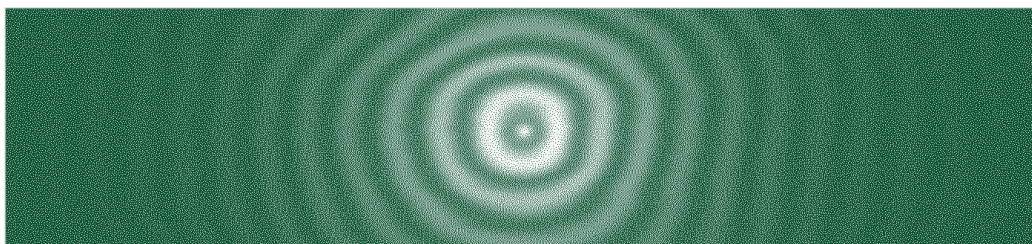
Lamb Wakefield

FLOCK CAPITAL · LAMB@FLOCK.HOLDINGS

vmfunc

FLOCK CAPITAL · CELESTE@LINUX.COM

*September 2, 2026*



ORBIT · SEED 1024 · FLOYD-STEINBERG ERROR DIFFUSION

## ABSTRACT

*Ideally, mechanistic interpretability would provide guarantees about the computational process used by a model: features should be causal, composable, and closed-form. Most existing techniques, such as sparse autoencoders, can perform arbitrarily badly on all three. Arguably the simplest interpretability technique is a linear probe, and the simplest control technique is a steering vector. They are remarkably robust compared to unsupervised techniques such as SAEs, but require labeled contrastive datasets. We propose a MoEfication technique that decomposes a representation space into a set of contrastive classifiers. Each classification corresponds to a representative vector; inputs are encoded as classifier outputs and reconstructed from a weighted average of the representative vectors. Steering effectively comes for free, outputs are bounded, the representative vectors admit static interpretation in principle, and feature space is structured. An evaluation against SAE baselines is reported: forced classifications steer causally, compose additively, and reach 0.43× the supervised steering effect on a shared frame, but reconstruction pays roughly 40× for the contrastive structure, head-level descriptions do not beat a null at the current sample size, and partitions do not reproduce across seeds. Training diagnostics, single-knob retrains, an HSIC-bottleneck ablation, a scaling sweep, and a reinforcement-learning probe of classification rewards are included.*

## 1 INTRODUCTION

Ideally we would like mechanistic interpretability to provide guarantees about the computational process used by a model. We would like features to be

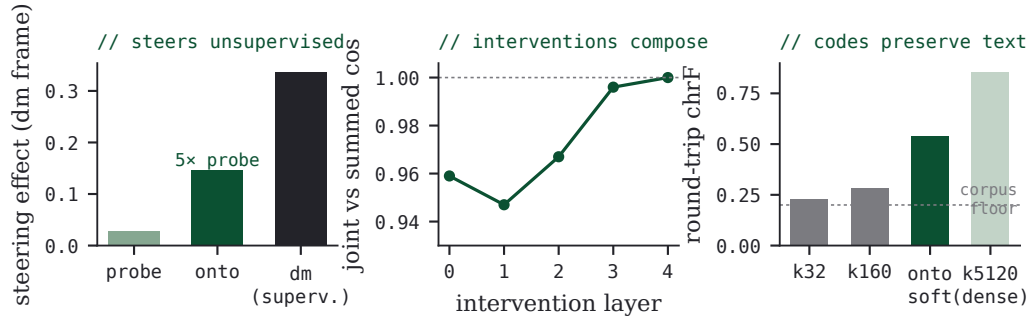
1. causal
2. composable
3. closed-form

However most existing techniques, such as sparse autoencoders (SAEs), produce features that can perform arbitrarily badly on these desiderata. Feature splitting breaks compositionality. They can't usually be statically analyzed (though bilinear SAE variants can be). Sparsity trades off against reconstruction fidelity, limiting the extent to which features can be used for steering.

Arguably the simplest interpretability technique is a linear probe, and the simplest control technique is a steering vector. They are remarkably robust compared to unsupervised techniques such as SAEs, but require labeled contrastive datasets. We propose a MoEfication technique based on decomposing a representation space into a set of contrastive classifiers. Each classification corresponds to a representative vector. Inputs are encoded as a series of classifier outputs, and reconstructed from a weighted average of the representative vectors. This has several advantages over SAEs:

1. It effectively gives steering for free.
2. Outputs are bounded.
3. The representative vectors admit static interpretation in principle; making that real is currently an open problem (Section 6.4).
4. Feature space is structured.

The measured version of this pitch is Figure 1; the full evaluation, including where it cuts against the proposal, is Section 6.



**FIGURE 1** · The proposition, measured. Left: forcing classifications moves decodes along the supervised difference-of-means direction  $0.43\times$  as far as supervised steering itself and  $5\times$  further than probe directions, the strongest unsupervised arm tested (Table 3). Center: joint interventions point where the summed single interventions point, exactly so in the final layer (working notes). Right: the sparse SAE tier’s codes lose most text before any intervention is applied (round-trip chrF  $0.23\text{--}0.29$  against a  $0.20$  corpus floor) while the Ontologizer’s soft code preserves it, so low-collateral steering is available only to the soft architectures; the dense k5120 SAE preserves text but is not a sparse steering code (working notes).

## 2 ASSUMPTIONS

This proposal assumes:

- **Steering is inherent to the training objective.** The dictionary vectors must also be high quality steering vectors; interpretability and steering are treated as the same problem.
- **Weights are themselves interpretable** (for certain architectural choices).
- **Clustering rather than decomposition.** Representations are grouped into mutually contrastive categories rather than decomposed into independently firing features. The clustering framing is counterintuitive unless you arrive at it from Leiden-style community detection.
- **Categorical structure is imposed on features.**
- **Interpretability gets a cheap forward pass.**
- **Concepts compose linearly.** Concepts must be additive to be useful.
- **Dictionary learning is all you need**, in a nonstandard form: the dictionary forms a basis for the representation rather than a target.
- **Representations are not themselves quantized.** The learned dictionary serves as a basis for the representation rather than a discrete codebook; quantization comes from sharpening of classifications over that basis.
- **Interpreting a well trained model is the same problem as interpreting the data and training tasks.**

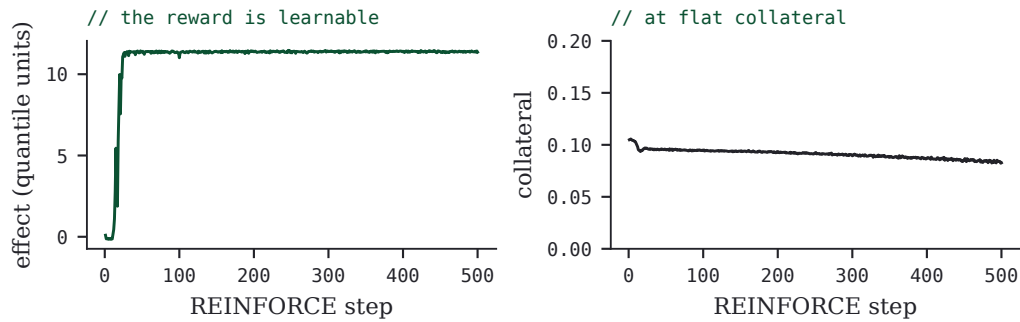


FIGURE 2 · REINFORCE against a single frozen classification’s quantile-normalized score, in-embedding tier. The policy saturates the reward within roughly 40 steps while collateral stays at its starting level; a classification is a trainable objective.

### 2.1 PIVOTAL ASSUMPTIONS

Two of these are pivotal: some version of the linear representation hypothesis is true, and latent space is in some way “natural”.

### 2.2 CONCERNS

**Timelines could be too short** We’ll just have to go faster.

**Contrastive classifiers might not be expressive enough** This is what the reconstruction benchmarks are for. The preliminary numbers (Section 6) put a price on the constraint; the open question is whether steering quality justifies it.

**Steering could push it too far off manifold** This is an open problem in mechinterp. Diffusion steering is one candidate to address it[21]. If we can’t solve the problem it still gives good data to test steering methods. e.g. can we RL a model for high/low scores on a classification and have it converge faster than paired steering or PCA vectors? A first pass exists: a REINFORCE policy over steering directions on frozen `resid_nc_hm`, rewarded by one classification’s quantile-normalized score at a matched intervention budget, converges cleanly in the differentiable in-embedding tier, saturating within roughly 40 steps at flat collateral (Figure 2). Neither the decode-cycle tier nor the convergence race against paired steering has been run; the race is the test this paragraph proposes.

**Representations could be too soft/dense/high entropy** This is highly correlated across all feature-based interp (SAEs, transcoders, J-space). Dividing representations across orthogonal categorical subspaces makes it possible to interpret the conceptual basis vectors even if the representations are dense mixtures.

**Ontofeatures could be too polysemantic** Orthogonality & categorical constraints give a theoretical inductive bias to monosemanticity. As long as they have decent explanatory power we can do eigendecomposition; polysemanticity decreases with parameter count and we have more directions to increase parameters (in addition to increasing width

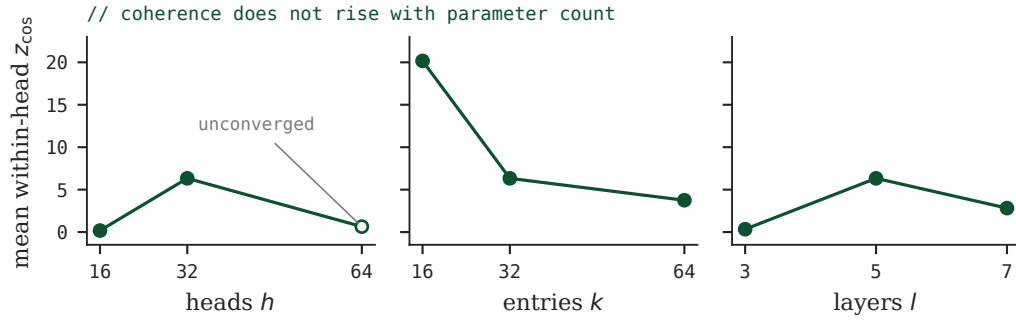


FIGURE 3 · Head coherence against each parameter axis of the star sweep, all runs at 4 epochs (62k steps) of the retrain recipe’s 24. No axis delivers the predicted rise; the entries axis falls monotonically. The open marker is the unconverged  $h=64$  run.

we can increase head size or stack more layers). A first star sweep tests the scaling prediction at reduced training (architecture and terms defined in Section 5.1): seven configurations around the shared  $h=32, k=32, l=5$  shape, varying one axis at a time, each trained 4 epochs (62k steps; the retrain recipe’s 24 epochs is 372k steps). No measured axis delivers the predicted rise: head coherence shows no positive relation to parameter count (Spearman +0.14 for mean  $z_{\cos}$ , -0.54 for the fraction above  $z=2$ ), usage imbalance grows with size (+0.86), and the cleanest per-axis trend runs opposite to the claim, coherence falling monotonically as entries per head grow ( $k=16/32/64$ : mean  $z_{\cos}$  20.2/6.3/3.7). One confound is recorded: the widest model ( $h=64$ ) has not converged at this budget (tail MSE 20× the others), so its collapse is not evidence (Figure 3). Seven points at short training are weak evidence either way; full-length retrains are the missing control.

**Interpreting ontofeatures could be otherwise intractable** This would imply the universality hypothesis is false and is highly correlated with intractability of other interpretability agendas. Submodules could be interpretable even if features aren’t.

**Training could be intractable** Sparsity trades against sample/compute efficiency; preliminary experiments suggest we have less need for a sparsity constraint (working notes).

**Too many hyperparameters/too hyperparameter dependent** This generally becomes less of an issue with scaling; preliminary experiments show most changes to hyperparameters just change the rate of convergence (working notes).

**Doesn’t converge in the limit** Collapse in one layer would propagate down the stack. Candidate causes: underparameterization, a really bad optimizer, bad initialization, a bad loss term.

**Not orthogonal in the limit**  $\text{cossim}_h$  (Section 5.2.2) penalises overlap between heads but does not guarantee disjoint support at convergence. The head geometry (Figure 6) is built to make orthogonality the cheap solution, not to force it.

**Might not generalize out of distribution** The same concern applies to every feature-based technique. Closed-form interpretation of the dictionary is the mitigation: a statically interpreted basis does not depend on the input distribution, though the classifications over it do.

**Inference too expensive** Each layer adds  $h \cdot k$  classifications per sample (Section 5.1), with  $k$  small; the code is computed without materializing a wide feature space. The cost is the extra decoder passes for deep supervision during training.

**Autointerp doesn't work on parameters/eigenfeatures** In the preliminary experiments, parameter-space decodes of feature directions failed as descriptions outright (Section 6.4). This is direct evidence against the static-interpretability claim.

### 3 MOTIVATION

#### 3.1 THREAT MODEL

*The ontology identification problem.* We can't currently look inside a model and make any conclusions about its goals or world model. This is a major limitation on almost all proposed alignment agendas.

**Inner misalignment** We don't currently know how to specify a **robust goal for a model**. Because goal space is large, we should expect **goal misgeneralization as the default outcome**<sup>1</sup>.

*Limits of natural abstractions.* To some extent, deep learning models seem to learn universal features irrespective of architecture[5]. This implies the existence of at least a weak version of natural abstractions[12], which would be a very convenient shortcut for auditing a model.

Natural abstractions could be considered the set of concepts we could use to communicate with aliens. They should consistently emerge through unsupervised learning (in the limit given some constraints). If any neural network is considered a fuzzy generalization of a finite state machine, natural abstractions are the latent states.

Unfortunately, the abstractions a model learns seem highly sensitive to choice of training data, and are thus not necessarily universal in the limit.<sup>2</sup> Assuming that model capabilities will continue to primarily be driven by access to more data and more modalities, we cannot expect the abstractions learned by a weak model to remain robust in a more powerful model.

Even if we could fully characterise a model's state space in terms of natural abstractions, that still doesn't mean we can reconstruct the latent FSM, as the transition rules can be arbitrarily complex.

<sup>1</sup>The most straightforward path to doom is for a goal-driven superintelligence to seize power to prevent its optimization target from being changed. In slow takeoff or multipolar scenarios, there is also a "decoherence" risk, where comparably weak specialist optimizers disempower humanity without any coherent strategy. Consider for instance superpersuaders which fall short of superintelligence. A viral chatbot might have very simple objectives to optimize engagement and distribution of itself. While such a scenario would be an especially **undignified** end for humanity, it broadly fits with both natural language being much easier than anyone expected and AI labs being much less safety conscious than anyone expected.

<sup>2</sup>The same applies to those that seem most natural to humans.

### 3.2 THEORY OF CHANGE

Conditional on the underlying model of ontology being true, the techniques proposed here would be a substantial interpretability advance. The agenda also has a case for counterfactual impact: its theoretical basis consists of nonobvious applications of obscure techniques from niche fields of bioinformatics, and similar research is unlikely to be done soon otherwise.

*What a solution would look like.* Mechinterp plausibly offers the best path to a robust alignment solution despite criticisms. The case rests on

1. **decomposability** Mechinterp allows alignment to be broken into manageable subproblems.
2. **parallelizability** Decomposable subproblems can be worked on independently.
3. **robustness** Mechinterp offers direct causal explanations of model behavior.

The most straightforward mechinterp alignment solutions are

1. **eliciting latent knowledge (ELK)** Directly inferring a model’s ontology.
2. **mechanistic anomaly detection (MAD)** Detecting when a model has “unusual” thoughts.

Both of these can be considered **ontology identification** subproblems. MAD requires mapping between the model’s activations and the model’s ontology. ELK requires mapping between the model’s ontology and human ontology. These are hard but not intractable problems. Recently, **developmental interpretability** (devinterp) techniques have enabled identification of “developmental stages” corresponding to learning specific tasks[9]. This is strong evidence that learning consists primarily of the network grokking[16] discrete concepts. While these results are promising for MAD, locating these emergent capabilities in the model’s activations remains an open problem.

Interpretability glosses broadly as “let humans understand AI”. The alternative is to get AI to understand humans better, and that scales badly: making the model smarter leads to subtler and subtler issues with larger and larger effects. And what we’re doing is not just letting humans understand AI. Because an LLM reflects humanity from its training data, high quality conceptual interpretability provides machine-comprehensible mathematical representations of every concept that humanity cares about. We work on interp for the usual corrigibility reasons, and also because feature representation is one of the most challenging parts of outer alignment. This attacks both.

*Macro-scale strategy.* Broadly, the goals are to

1. formalize the notion of a model’s latent concept space
2. distill a model’s latent concept space
3. enforce decomposability of latent concept space
4. enforce robustness of existing concepts as the dimensionality of the latent concept space increases

A prerequisite for all of these is the ability to express a **metaontology** which can identify latent concepts in activation space.

## 4 BACKGROUND

Mechanistic interpretability currently can do supervised steering well or unsupervised steering poorly. This has prompted the Google Deep Mind interpretability team to pivot to a pragmatic steering agenda[15]. Compared to unsupervised interpretability, supervised steering is highly robust. However it can only distinguish what the experimenter is looking for and does not account for the actual computations performed by a model[13]. The dominant unsupervised paradigm is the sparse autoencoder (SAE). A model’s hidden state is approximated as a finite state machine consisting of discrete features. The initial implementation used a ReLU activation and sparsity loss[1], though more recent implementations have used a TopK activation function without a sparsity loss[8].

### 4.1 COMMON PROBLEMS WITH SAEs

The motivation is to decompile a model to a finite state machine (FSM). The limitation is that such an FSM is only as interpretable as an enumeration of all possible states.

*Features are not composable.* A persistent problem with SAEs is feature absorption[4]. Instead of representing an input as a composition of monosemantic features, an SAE will often instead use a single separate feature. This is a problem for interpretability, as we would like features to generalize. One attempt to address this problem is Matryoshka SAEs[2], which use multiple layers to build a hierarchical coarse-to-fine representation.

*Features are not causal.* Sparsity trades off against reconstruction fidelity. Using features for steering often results in inconsistent, context-sensitive behavior[6].

*Features are not closed-form.* A major concern is how well an interpretability technique generalizes out-of-distribution. Ideally we would like to infer behavior by looking solely at model weights, which can be achieved with bilinear MLPs[19].

*Forward passes are expensive.* Efficiency is also a problem. Although the sparsity constraint means that only a handful of features are active for a given input, the entire feature space needs to be computed for each sample. Mixture-of-experts (MoE) SAE variants reduce compute costs by dividing feature space between smaller submodules. Examples include Switch SAEs[14] and Tree SAEs[3].

Less explored is the effect of the structure imposed by the MoEfied architecture on the interpretability of feature space. A common approach to scalably interpreting features is using an LLM to predict activations from examples[18]. With an unstructured feature space, it quickly becomes intractable to interpret all permutations of features this way. It is however possible to instead apply this technique to expert routings. The MLPs of a transformer can themselves be converted to an MoE architecture[23]. Language models seem to organize contrastive concepts into orthogonal simplices[17]. This is suggestive of MoEfication revealing latent contrastive steering vectors. Imposing an orthogonality constraint has been shown to significantly reduce feature absorption[11].

Existing attempts to route activations to expert SAEs focus on routing similar inputs to the same expert. Contrastive learning wants the opposite: training on things at opposite ends of a shared axis. We don’t just want to find similar features, we want to find contrastive vectors in feature space that correspond to what a model considers a natural category.

## 5 METHODS

### 5.1 ARCHITECTURE

We propose a novel method which encodes samples as classifications mapped to contrastive vectors. This directly constrains a model’s representations. Moreover, classifications can natively be used for steering.

The model (Figure 4) consists of  $l$  DictEnc layers writing into a shared residual accumulator, decoded by a single linear map. Each layer (Figure 5) is composed of a classifier layer followed by a dictionary layer. The classifier splits the input into  $h$  independent soft label vectors. The dictionary maps these labels to representative vectors. Each layer writes a weighted average of these vectors to an accumulator. At each layer, the accumulator  $R_i$  is decoded and the residual  $X - \text{dec}(R_i)$  is passed to the next layer. Deep supervision is used to constrain each layer to reconstructing the input.

Because labels within a head are mutually exclusive rather than independent, each dictionary head serves as a contrastive set. An auxiliary loss term minimizing  $\text{cossim}_h$  (Eq. 6) pushes each head to represent an orthogonal subspace (Figure 6). Because the dictionary returns a linear combination of representative vectors, subspaces are convex.

*DictEnc.* Each DictEnc consists of a multihead bilinear classifier (BilinearBlock) with weights  $[W, V] \in \mathbb{R}^{2 \times h \times k \times d}$  ( $d + 1$  after the first layer, see Eq. 14) and a multihead linear dictionary (DictBlock) with weights  $D \in \mathbb{R}^{h \times k \times e}$ . It accepts a vector  $\mathbf{x} \in \mathbb{R}^d$ , which may be a residual from a previous layer.

*Classifier.* A BilinearBlock accepts inputs  $\mathbf{x} \in \mathbb{R}^d$ . It returns logits  $K \in \mathbb{R}^{h \times k}$  given by

$$\text{BilinearBlock}([W, V], \mathbf{x}) := \left( \sum_{j=1}^d W_{\cdot, \cdot, j} x_j \right) \odot \left( \sum_{j=1}^d V_{\cdot, \cdot, j} x_j \right) \quad (1)$$

An optional noise term  $\varepsilon_K$  may be added. Winner dropout sets  $\arg\max_k(K_{h, \cdot})$  to  $-\infty$  with probability  $p_{\text{drop}}$ , annealed from  $0 \rightarrow 0.1$ .

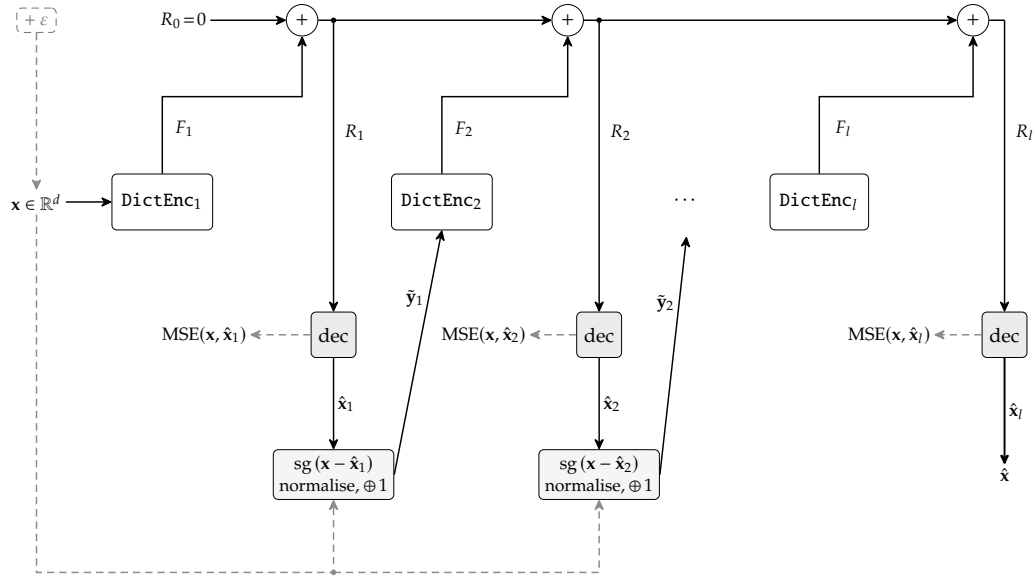
*Labels.* Soft labels  $P$  are given by  $\text{softmax}_k(K/t)$  where  $t$  is a temperature hyperparameter. It is annealed geometrically from 1.0 to 0.03 over the first 50k steps, then held, giving progressively sharper classifications. The architecture provides methods for causal interventions on labels. From  $P$  we collect per-sample entropy  $H$ , in bits, so a head saturates at  $\log_2 k = 5$  bits,

$$\text{entropy}_k(P)_i := - \sum_{j=1}^k P_{i,j} \log_2 P_{i,j}, \quad H := \mathbb{E}_b \mathbb{E}_h [\text{entropy}_k(P)] \quad (2)$$

between-sample similarity  $\text{cossim}_b$ , the mean absolute cosine similarity of the per-head features  $F$  over all ordered pairs of samples,

$$\text{cossim}_b(P) := \mathbb{E}_{n,q} \left[ \left\| \frac{\langle \mathbf{f}_n, \mathbf{f}_q \rangle}{\|\mathbf{f}_n\| \|\mathbf{f}_q\|} \right\| \right], \quad \langle \mathbf{f}_n, \mathbf{f}_q \rangle = \sum_{i=1}^h \mathbf{p}_{n,i}^\top G_i \mathbf{p}_{q,i} \quad (3)$$

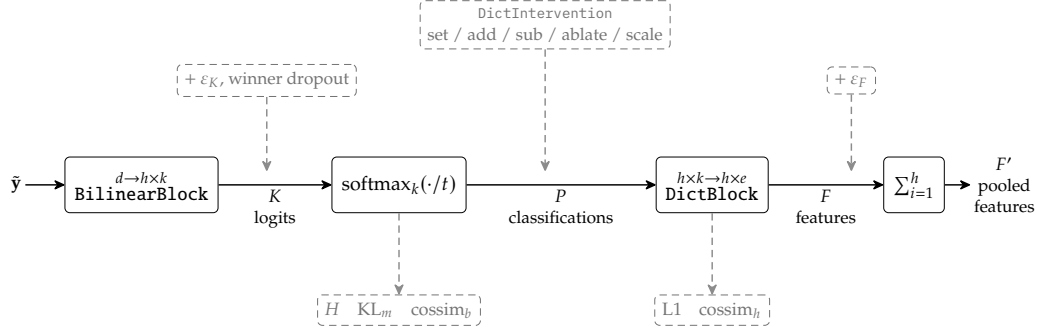
where  $\mathbf{f}_{n,i} := F_{n,i} \in \mathbb{R}^e$  is head  $i$ ’s feature vector for sample  $n$ ,  $\mathbf{f}_n = \bigoplus_{i=1}^h \mathbf{f}_{n,i}$  is their



**FIGURE 4 · Ontologizer architecture.** An Ontologizer consists of  $l$  stacked DictEnc layers. Each returns a feature vector  $F_i \in \mathbb{R}^e$ , which is written to a residual accumulator  $R$ . For each layer  $i$ ,  $R_i = R_{i-1} + F_i$ ; a stop-gradient variant (sg on  $R_{i-1}$ , available but off in the live configuration) would prevent gradient communication between layers. dec is a linear decoder with shape  $\mathbb{R}^{d \times e}$ . It uses deep supervision to obtain a coarse-to-fine representation where the component encoded by previous layers is inaccessible to later layers. At each layer  $i$ ,  $\hat{x}_i = \text{dec}(R_i)$ . The same decoder is shared by all layers. The component left unexplained after layer  $i$  is given by  $y_i = x - \hat{x}_i$ . The input to layer  $i + 1$  is given by  $\tilde{y}_i = y_i / \text{L2}(y_i) \oplus [1]$  where L2 is the unit norm. This prevents exploding gradients from repeated application of bilinear layers, as each scales its input by  $\|\cdot\|^2$ . A constant coordinate  $[1]$  is appended, giving a dimension of  $d + 1$ . This prevents the sign information of  $y_i$  from being lost by the bilinear transformation. Effectively, it adds a bias to an otherwise unbiased linear form while preserving eigendecomposition.

$\text{MSE}(x, \text{dec}(R_i))$  is computed separately for each layer.

The input-noise hook drawn on  $x$  is available but disabled in the live configuration.



**FIGURE 5 · DictEnc architecture.** Dashed lines above indicate perturbations to the data, including noising and causal interventions. Dashed lines below indicate statistics used as tuneable auxiliary loss functions. Each DictEnc consists of a multihead bilinear classifier (see §5.1.2) and a multihead linear dictionary (see §5.1.4). It accepts a vector  $\mathbf{x} \in \mathbb{R}^d$ , which may be a residual from a previous layer. Optional noise terms  $\varepsilon_K$  and  $\varepsilon_F$  may be added (see §5.1.6). Winner dropout sets  $\text{argmax}_k(K_{h,\cdot})$  to  $-\infty$  with probability  $p_{\text{drop}}$ , ramped  $0 \rightarrow 0.1$  over the annealing horizon. Soft labels  $P$  are given by  $\text{softmax}_k(K/t)$  (see §5.1.3). The architecture provides methods for causal interventions on labels. From  $P$  we collect per-sample entropy  $H$  (Eq. 2), between-sample similarity  $\text{cossim}_b$  (Eq. 3), and batch mean entropy  $\text{KL}_m$  (Eq. 4). A DictBlock accepts labels  $P \in \mathbb{R}^{h \times k}$  and returns features  $F \in \mathbb{R}^{h \times e}$  (Eq. 5). From  $F$  we collect between-head similarity  $\text{cossim}_h$  (Eq. 6) on the clean features, and the sparsity penalty L1 (Eq. 7).  $F$  is summed over the head dimension to obtain pooled features  $F' \in \mathbb{R}^e$ , which are added to  $R$ .

concatenation, and  $G_i = |D_i||D_i|^\top \in \mathbb{R}^{k \times k}$  is head  $i$ 's dictionary Gram. The implementation evaluates Equation 3 in tag space through  $G$  rather than materializing  $F$ ; the two are identical in value and gradient, but  $O(hk^2e + b^2hk)$  instead of  $O(b^2he)$ .

Finally,  $\text{KL}_m$  is the divergence of the batch-mean label vector from uniform, in bits, averaged over heads.

$$\text{KL}_m(P) := \mathbb{E}_h [\log_2 k - \text{entropy}_k(\mathbb{E}_b[P])] \quad (4)$$

In practice  $H$  is not needed, as including a  $\text{cossim}_b$  loss also minimizes  $H$ .  $\text{KL}_m$  serves to anchor the batch mean of each label vector  $\mathbf{p}_{i,\cdot}$  to a uniform distribution. This results in a uniform  $\mathbf{p}_{i,\cdot}$  code corresponding to a maximally generic feature vector.

*Dictionary.* A DictBlock accepts labels  $P \in \mathbb{R}^{h \times k}$  and returns features  $F \in \mathbb{R}^{h \times e}$  given by

$$\text{DictBlock}(D, P) := \sum_{j=1}^k |D_{\cdot,j}| P_{\cdot,j} \quad (5)$$

Because  $|D_{\cdot,j}|$  is an absolute value and  $P_{hj} \in [0, 1]$ , features are strictly additive. From  $F$  we collect between-head similarity  $\text{cossim}_h$  on the clean features — the mean absolute cosine similarity over all ordered pairs of heads *within* a sample —

$$\text{cossim}_h(F) := \mathbb{E}_b \mathbb{E}_{i,j} \left[ \left| \frac{\langle \mathbf{f}_{n,i}, \mathbf{f}_{n,j} \rangle}{\|\mathbf{f}_{n,i}\| \|\mathbf{f}_{n,j}\|} \right| \right] \quad (6)$$

and the sparsity penalty L1 after the noise  $\varepsilon_F$  is added,  $\tilde{F} := F + \varepsilon_F$ . L1 is a sum, not a mean, so it scales with  $b$ ,  $h$ , and  $e$ ; its loss weight is correspondingly small.

$$L1(\tilde{F}) := \sum_{n=1}^b \sum_{i=1}^h \sum_{j=1}^e |\tilde{F}_{n,i,j}| \quad (7)$$

Both  $L1$  and  $\text{cossim}_h$  tend to decrease with only MSE loss, but incorporating them into the loss speeds up convergence.

**Disjoint support**  $\text{cossim}_h$  is of particular interest because on nonnegative features it is not a rotation constraint but a support-partition constraint. Every  $\mathbf{f}_{n,i}$  lies in the nonnegative orthant of  $\mathbb{R}^e$ : the dictionary contributes  $|D| \geq 0$  and the labels  $P \geq 0$ . For nonnegative  $\mathbf{u}, \mathbf{v}$  the inner product  $\langle \mathbf{u}, \mathbf{v} \rangle = \sum_j u_j v_j$  is a sum of nonnegative terms, so it vanishes if and only if every term does,

$$\langle \mathbf{u}, \mathbf{v} \rangle = 0 \iff \text{supp}(\mathbf{u}) \cap \text{supp}(\mathbf{v}) = \emptyset \quad (8)$$

Signed vectors admit no such conclusion: there orthogonality is cancellation, and two vectors with identical support can be orthogonal. This is why  $\text{cossim}_h$  ranges over  $[0, 1]$  rather than  $[-1, 1]$ , and why the absolute value in Equation 6 is vacuous for  $\text{cossim}_h$  though not for  $\text{cossim}_b$ . It is also why the statistic is collected before  $\varepsilon_F$  is added: noised features leave the orthant and the argument with it.

The constraint binds the dictionary, not merely one sample's features. Because  $P$  is a softmax,  $p_{i,a} > 0$  for every entry, and expanding Equation 5 gives

$$\langle \mathbf{f}_{n,i}, \mathbf{f}_{n,j} \rangle = \sum_{a=1}^k \sum_{b=1}^k p_{i,a} p_{j,b} \langle |\mathbf{d}_{i,a,\cdot}|, |\mathbf{d}_{j,b,\cdot}| \rangle \quad (9)$$

a strictly positive combination of all  $k^2$  cross-head entry overlaps. A single sample on which heads  $i$  and  $j$  are orthogonal therefore forces  $\langle |\mathbf{d}_{i,a,\cdot}|, |\mathbf{d}_{j,b,\cdot}| \rangle = 0$  for every pair of entries  $(a, b)$ : the two heads own disjoint blocks of coordinates as a property of  $D$ , holding for all inputs rather than on the data that happened to be measured. That is the premise a static reading of the dictionary needs, and it is stronger than what any signed-feature architecture can ask for. The budget is not tight either. At the live configuration  $h = 32$  heads partitioning  $e = 2048$  coordinates leaves 64 dimensions per head to hold  $k = 32$  entries, which need not be disjoint from each other within a head; head-disjointness is most of what the width  $e > d$  buys.

Two different orthogonalities are in play and only the between-head one is architectural. Disjoint support assigns each head a block of coordinates; *within* a head the  $k$  entries are contrastive alternatives spread apart inside that block (Figure 6). Section 6.2 measures the second. Neither is guaranteed at convergence:  $\text{cossim}_h$  is a penalty, and the geometry is built to make a support partition the cheap solution rather than to force one.

We add noise  $\varepsilon_F$  to  $F$ , then sum over the head dimension to obtain pooled features  $F' \in \mathbb{R}^e$ , which are added to  $R$ .

$$F' = \sum_{i=1}^h \tilde{F}_{i,\cdot} \quad (10)$$

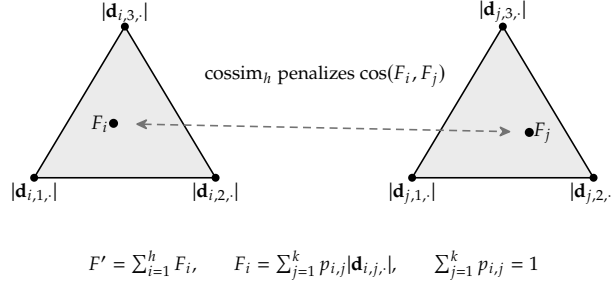


FIGURE 6 · DictBlock **head geometry** for  $h = 2; k = 3$ . The output space of a head  $i$  forms a simplex with the rows of the matrix  $D_{i,·,·}$  as the vertices. The row vectors  $\mathbf{d}_{i,j,·}$  form the spanning set of the feature space. Any  $F_i$  must be a linear combination of the rows of  $D_{i,·,·}$ . As  $F$  becomes sparser, these vectors become more approximately orthogonal, allowing them to be treated as a basis. The label code  $P$  corresponds to soft assignments to mutually contrastive categories rather than independently firing features. Because elements of  $P$ ,  $p_{ij} \in [0, 1]$ , outputs must be within the convex hull of  $D_{i,·,·}$ . Because  $|D| \geq 0$  and  $P \geq 0, F \geq 0$ ; therefore  $\cos(F_i, F_j) \in [0, 1]$ , and driving  $\text{cossim}_h(F)$  to zero asks the heads to partition the coordinates of  $\mathbb{R}^e$  between them (Section 5.1.4).

*Decoder.* Each DictEnc output  $F'_i$  is added to an accumulator  $R$ .

$$R_i = \sum_{j=1}^i F'_j \quad (11)$$

A single-layer unbiased linear decoder  $\text{dec}$  projects  $R_i$  back to the input space  $\mathbb{R}^d$ . Weights  $W_{\text{dec}} \in \mathbb{R}^{d \times e}$  are shared for all DictEnc $_i$ .

$$\hat{X}_i = W_{\text{dec}} R_i \quad (12)$$

The residual  $Y_i$  is given by  $X - \hat{X}_i$ , stop-graduated so that lower layers cannot write a communication code into the residual instead of reducing it. It is normalized by the unit norm L2, with  $\epsilon$  the floating-point epsilon of the working dtype.

$$Y_i = \text{sg}[X - \hat{X}_i], \quad \text{L2}(\mathbf{y}) := \sqrt{\sum_{j=1}^d y_j^2} \quad (13)$$

$$\tilde{\mathbf{y}}_i = \frac{\mathbf{y}_i}{\text{L2}(\mathbf{y}_i) + \epsilon} \oplus [1] \quad (14)$$

A constant coordinate  $[1]$  is appended, giving a dimension of  $d + 1$ . This prevents the sign information of  $\mathbf{y}_i$  from being lost by the bilinear transformation. Effectively, it adds a bias to an otherwise unbiased linear form while preserving eigendecomposition.

*Noise.* Noise is added to  $X$ ,  $K$ , and  $F$ , controlled by hyperparameters  $\sigma_X$ ,  $\sigma_K$ , and  $\sigma_F$ . Each draws a standard normal of the same shape as its target,

$$\epsilon \sim \mathcal{N}(0, 1) \quad (15)$$

and scales it three different ways. Classifier logits take unscaled Gaussian noise,

$$\text{noise}_{\text{normal}}(K) := K + \sigma_K \epsilon \quad (16)$$

inputs take noise proportional to their own norm, so the scale is a fraction of signal magnitude rather than an absolute quantity,

$$\text{noise}_{\text{batchnorm}}(X) := X + \sigma_X \varepsilon \odot \frac{\text{sg}[\text{L2}(X)]}{\sqrt{d}} \quad (17)$$

and features take noise proportional to each feature’s standard deviation across the batch.

$$\text{noise}_{\text{featvar}}(\mathbf{F}) := \mathbf{F} + \sigma_F \varepsilon \odot \text{sg} \left[ \sqrt{\text{var}_b(\mathbf{F}) + \epsilon} \right] \quad (18)$$

Here  $\text{L2}(X)$  is the per-sample norm of Equation 13, broadcast over  $d$ . Both scales are stop-gradiented: without  $\text{sg}$  the model could evade its own noise by shrinking feature variance, and the  $\epsilon$  inside the root guards the 0/0 gradient where a feature’s batch variance underflows to exactly zero. Live values are  $\sigma_X = 0$  (input noise off),  $\sigma_K = 0.02$  annealed geometrically to  $0.2 t_{\text{end}}$ , and  $\sigma_F = 0.1$ . Because  $\sigma_K$  enters the softmax as  $\sigma_K/t$ , holding it constant under the falling temperature would grow effective exploration rather than hold it.

## 5.2 TRAINING

*Embedding model.* We embed texts using SONAR[7], a multimodal and language-agnostic sentence embedding model. SONAR embeddings have several desirable properties:

- They correspond to entire texts rather than tokens.
- They are bidirectional.
- Language is encoded independently of text.

This allows us to validate steering fidelity against a ground truth. Large Concept Models[20] generate directly in SONAR space, which makes it a natural substrate for steering experiments.

*Loss function.* Reconstruction fidelity is trained with MSE, where  $\mathbf{w} \in \mathbb{R}^d$  is a fixed per-dimension weight vector.

$$\text{MSE}(X, \hat{X}) := \frac{1}{bd} \sum_{n=1}^b \sum_{j=1}^d w_j (\hat{x}_{n,j} - x_{n,j})^2 \quad (19)$$

The total objective adds the ghost-gradient reconstruction and the auxiliary statistics, each weighted by a scalar. Statistics are *summed* over the  $l$  layers rather than averaged.

$$L := \text{MSE}(X, \hat{X}) + s_g \text{MSE}(\hat{X}_g, X - \hat{X}) + \sum_{i=1}^l \mathbf{s} \cdot \mathbf{m}_i \quad (20)$$

$$\mathbf{m}_i = \left[ \text{L1}(K_i) \quad \text{L1}(\tilde{\mathbf{F}}_i) \quad H_i \quad \text{cossim}_b^{(i)} \quad \text{cossim}_h^{(i)} \quad \text{KL}_m^{(i)} \right]^\top \quad (21)$$

Live weights are  $s_{\text{L1F}} = 10^{-9}$ ,  $s_{\text{cossim}_b} = 10^{-5}$ ,  $s_{\text{cossim}_h} = 10^{-6}$  and  $s_{\text{KL}_m} = 10^{-6}$ , with  $s_{\text{L1K}} = s_H = 0$ ; the ghost path is disabled, since  $\hat{X}_g$  fires only on features that are exactly zero across the whole batch, which bilinear classifiers with softmax selection and absolute-valued dictionaries never produce.

Reconstruction is trained with deep supervision:  $\text{MSE}(X, \text{dec}(R_i))$  is computed separately at every layer and averaged, so each prefix carries weight  $1/l$  and each layer must explain what previous layers left unexplained. The MSE is whitened:  $\mathbf{w}$  in Equation 19 is the per-dimension inverse variance, normalized to mean one, the same whitening the evaluation applies. The implementation applies  $\sqrt{\mathbf{w}}$  to  $X$  and  $\text{dec}(R_i)$  before the squared error, which is equivalent. Four auxiliary statistics are active as loss terms, of six the implementation exposes (Figure 5): between-sample similarity  $\text{cossim}_b$ , computed from  $P$  through the dictionary Gram so it measures feature-space similarity of the per-head reconstructions, and which in practice also minimizes per-sample entropy  $H$ ; the batch-mean term  $\text{KL}_m$ , anchoring each head’s mean label vector to a uniform distribution so that a uniform code corresponds to a maximally generic feature; an L1 penalty on features; and between-head similarity  $\text{cossim}_h$ , which measures the extent to which heads encode nonoverlapping geometries. L1 and  $\text{cossim}_h$  tend to fall under MSE alone; including them in the loss speeds up convergence. Classification sharpness is controlled by annealing the softmax temperature geometrically from 1.0 to 0.03 over the first 50k steps, then holding it. Winner dropout and the noise terms  $\varepsilon_K$  and  $\varepsilon_F$  regularize classifications and features respectively.

### 5.3 EVALUATION

Evaluation runs against top- $k$  SAEs trained on identical cached SONAR embeddings under an identical whitened reconstruction objective. The battery is deliberately SAEBenchmark[10]: a reconstruction-versus-capacity Pareto; categorical-form and semantic-coherence tests of head structure; cross-model feature splitting and seed stability; downstream text fidelity through the paired M2M100 decoder; sparse language probing; shrinkage decomposition; steering fidelity; and blinded auto-interpretability scoring. A structural-ablation ladder of SAE variants, each adding one Ontologizer-style commitment at a time, attributes effects to specific structure. AxBench[22] found that simple supervised baselines beat SAEs at steering; the steering comparison here is built to face that result.

## 6 PRELIMINARY RESULTS

The evaluation is in progress. Results so far are mixed. The full methodology and numbers are in the working notes<sup>3</sup>; the headlines are summarized here. Two models recur throughout and are named once now. The notes’ reference model, `resid_nc` (372,600 steps), is the model behind the notes’ tables, including Tables 1 and 2. Its sibling `resid_nc_hm` is the same configuration plus the batch mean-entropy bonus the notes propose as the freeze fix, trained to 4.38M steps; the new experiments reported here use its final checkpoint unless stated otherwise.

The notes carry one caveat about the reference: “layer 4 froze 22/32 heads late in temperature annealing and is mildly harmful.” A diagnostic over both training histories (frozen: top classification share  $\geq 0.95$  and churn  $\leq 0.02$ ) bears the caveat out and locates the fix (Figure 7). In `resid_nc`, the top layer (layer 4 of 0–4) begins freezing near step 190k, well after the anneal, reaches 27 of 32 by step 210k and 28 at its peak, and is still frozen at the last checkpoint; the count differs from the notes’ 22 under this criterion, the layer and the persistence do not. In `resid_nc_hm` the late freeze never occurs: what remains is

<sup>3</sup>K. Wiechecki, *Dictionary learning as unsupervised steering-vector generation: Ontologizers vs. sparse autoencoders on SONAR sentence embeddings*, working notes, in preparation; distributed with this proposal.

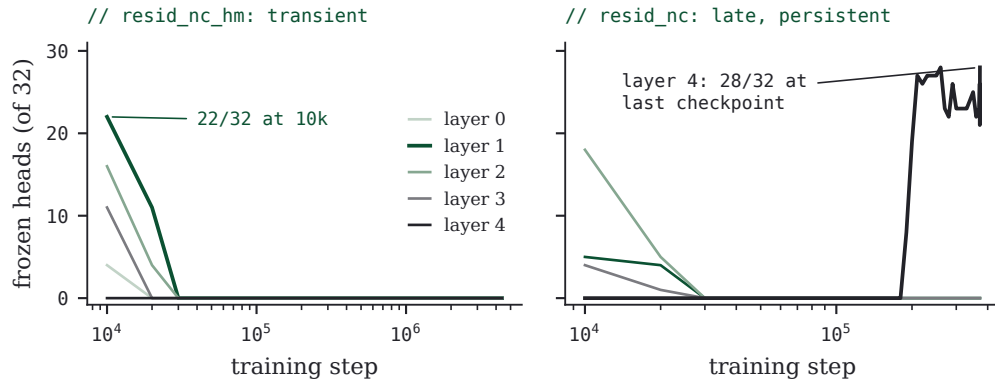


FIGURE 7 · Frozen heads per layer over training (frozen: top classification share  $\geq 0.95$ , churn  $\leq 0.02$ ). Left: `resid_nc_hm`, with the mean-entropy bonus, has only an early annealing transient, peaking at 22/32 in layer 1 at step 10k and recovering to zero everywhere by step 30k. Right: `resid_nc`, the notes’ reference, carries the caveat: a post-anneal layer-4 freeze from  $\sim 190k$  that persists through its last checkpoint.

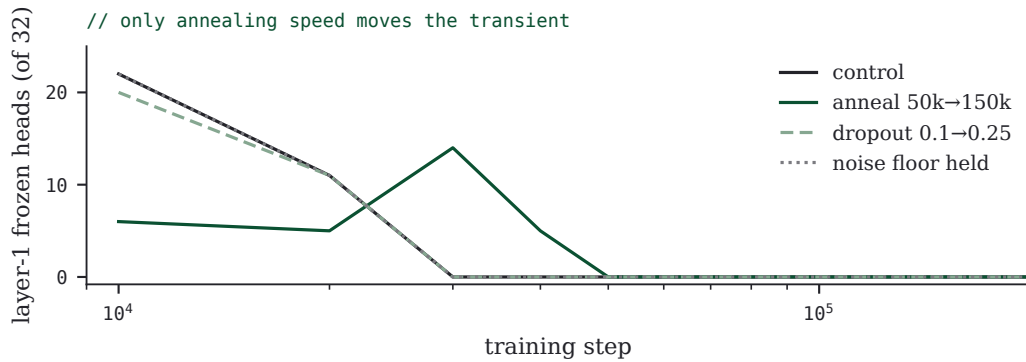


FIGURE 8 · The early transient under the four retrains, layer-1 frozen count. The noise-floor variant overlaps the control exactly and the dropout variant tracks it to within two heads at the peak; only stretching the temperature anneal moves the curve, trading peak height for duration.

an early annealing transient (22 of 32 at layer 1 of 0–4 at step 10k, with 16, 11, and 4 of 32 at layers 2, 3, and 0) that recovers completely by step 30k, leaving the final checkpoint unfrozen everywhere. The mean-entropy bonus does what the notes proposed it for.

Four retrains of the `resid_nc_hm` configuration, a zero-change control and three single-knob variants (372k steps each), close the question of what drives the remaining transient. The control reproduces it head-for-head (22/32 at layer 1 at step 10k, zero everywhere by 30k) and shows no late freeze over its horizon. Stretching the temperature anneal from 50k to 150k steps attenuates and spreads the transient (peak 14/32 at layer 1 at step 30k, clear by 50k); raising winner dropout from 0.1 to 0.25 tracks the control to within two heads at the peak, and holding the classification-noise floor at its starting value changes nothing at all (Figure 8). The transient belongs to the temperature schedule alone, and every retrain ends unfrozen everywhere.

| point                                          | coeffs | index bits | FVU <sub>w</sub> |
|------------------------------------------------|--------|------------|------------------|
| onto $m=0$ ( $\mathbb{E}[p]$ origin)           | 0      | 0          | 1.001            |
| onto hard (argmax)                             | 0      | 800        | 17.34            |
| sae k32 tier ( $m=5120/11264$ , incl. p5, s43) | 32     | ~400       | 0.494–0.509      |
| sae k32 bilinear                               | 32     | 394        | 0.602            |
| onto dev $m=1$                                 | 160    | 800        | 0.677            |
| sae g160top1                                   | 160    | 1972       | 0.274            |
| sae k160 ( $m=11264$ )                         | 160    | 2154       | 0.258            |
| onto dev $m=2$                                 | 320    | 1600       | 0.497            |
| onto dev $m=4$                                 | 640    | 3200       | 0.289            |
| onto dev $m=8$                                 | 1280   | 6400       | 0.094            |
| sae k5120 ( $m=11264$ , $L_0 \approx 1520$ )   | 1520   | 20455      | <b>0.00011</b>   |
| onto dev $m=16$                                | 2560   | 12800      | 0.0082           |
| sae g160softmax                                | 5120   | 0          | <b>0.00436</b>   |
| onto soft ( $m=32$ )                           | 5120   | 0          | 0.00538          |

TABLE 1 · Reconstruction against code capacity on the shared evaluation tail, reproduced from the working notes. Ontologizer points are top- $m$  deviation truncations around the measured  $\mathbb{E}[p]$  origin; SAE points use measured eval  $L_0$ . The plain-SAE frontier dominates the Ontologizer’s deviation code at every measured capacity; the gap is the price of contrast-set structure.

### 6.1 RECONSTRUCTION QUALITY

Contrast-set structure is expensive. The plain-SAE frontier dominates the Ontologizer’s deviation code at every capacity tested (Table 1), and the grouped-simplex constraint alone costs roughly 40× in weighted fraction of variance unexplained. Here FVU<sub>w</sub> is whitened MSE over the whitened variance of a constant predictor, and the deviation code is a per-head top- $m$  truncation around the measured mean classification  $\mathbb{E}[p]$ . The train/eval asymmetry carries over from the notes: the Ontologizer trained on the full cache including the evaluation tail, while the SAEs excluded it; a fresh-encoded evaluation set is blocked on a data source (Section 7.1).

### 6.2 ORTHOGONALITY

Trained heads carry the signature the steering goal predicts: within-head decode directions repel, with 96% of Ontologizer heads below a random-group null (Table 2). Heads form well-separated contrast sets rather than semantic clusters, in both the Ontologizer and its single-layer grouped-softmax analogue. This is the within-head axis: it says the  $k$  entries of a head spread apart, not that distinct heads occupy disjoint coordinates. The between-head disjoint support of Section 5.1.4 is a separate claim and is not tested here.

### 6.3 CATEGORICAL STRUCTURE

Categorical form, exhaustive and exclusive, exists precisely when it is trained in as hard selection. It never emerges spontaneously, and it is not recoverable post hoc.

### 6.4 AUTOMATED INTERPRETABILITY

Per-latent evaluation lenses systematically mis-measure classifier codes, whose content lives in partitions rather than latent magnitudes. Blinded descriptability tracks code

| grouping                    | groups | mean $z_{\cos}$ | frac $z > 2$ | frac $z < -2$ |
|-----------------------------|--------|-----------------|--------------|---------------|
| g160top1 (trained, hard)    | 160    | +161.3          | 1.00         | 0.00          |
| k32 discovered (m11264)     | 359    | +1.7            | 0.36         | 0.02          |
| onto heads (trained, soft)  | 160    | -17.6           | 0.04         | <b>0.96</b>   |
| g160softmax (trained, soft) | 160    | -14.7           | 0.00         | <b>1.00</b>   |

TABLE 2 · Within-head decode-direction coherence, z-scored against size-matched same-pool nulls ( $n_{\text{null}}=500$ ), reproduced from the working notes. Softmax-trained heads are anti-coherent: in 96–100% of heads the entries’ decode directions repel, so heads hold well-separated contrastive alternatives rather than semantic clusters. Hard per-group selection shows the opposite signature.

density rather than architecture, and parameter-space decodes of feature directions fail as descriptions outright.

A head-level pass over `resid_nc` (step 372,600) tests this directly; its frozen top layer is a caveat for any layer-4 heads in the sample. Decoding descriptions from dictionary parameters fails at the partition level as it does per latent (median self accuracy 0.02). Activation-grounded head descriptions do no better than a null pairing at the current sample size ( $n=20$  heads per condition, median accuracy 0.083 against 0.125), and a head’s detection accuracy is uncorrelated with the mean describability of its latents (Spearman +0.07 over 30 head–lens pairs, 15 heads under each lens; neither per-lens value distinguishable from zero). The two lenses disagree and neither shows above-chance head-level signal, so the mis-measurement argument predicts an advantage the current sample fails to deliver.

#### 6.5 STEERING FIDELITY

There is a causally verified existence proof: forcing a single classification in the first layer reliably steers decodes into Burmese while otherwise preserving content. Joint interventions compose approximately additively (median cosine between summed and joint decode deltas at least 0.95, exact in the final layer). Text fidelity for soft codes sits near the likelihood ceiling, while the sparse SAE tier starts at roughly 0.75 collateral, so low-collateral steering is currently exclusive to the soft architectures.

The matched-budget overlay against supervised baselines is now in (Table 3): difference-of-means, linear-probe, and forced-classification arms with a random-direction floor, medians over 24 language-magnitude cells per arm on language steering. Each arm wins its own effect frame, as expected. On the shared difference-of-means frame, classification forcing moves decodes along the target direction 0.43× as far as supervised steering does, and lands well ahead of probe directions. The collateral column is the check on all of it: every arm’s collateral, supervised steering included, sits at the random-direction floor (0.72–0.78), and none materially beats it. On task naturalness the arms are so far indistinguishable (Figure 9): chrF against unsteered decodes falls from ~0.35 at magnitude 0.25 to ~0.15 at magnitude 1.0 for the difference-of-means and forced-classification arms, with probe and random decaying in parallel from ~0.38 to ~0.20, and forced classifications tracking difference-of-means within noise throughout. A blinded human-rating sheet (120 items) is prepared but unscored.

The decisive comparison, though, is architectural. A single-layer grouped-softmax SAE reproduces the Ontologizer’s code structure, contrastive signature, and text fidelity at better reconstruction with half the parameters. It is the rival null hypothesis the full

| arm                    | eff (native) | eff (dm frame) | hit rate | collateral |
|------------------------|--------------|----------------|----------|------------|
| difference of means    | 0.337        | 0.337          | 0.250    | 0.757      |
| classification forcing | 0.394        | 0.146          | 0.125    | 0.778      |
| linear probe           | 0.293        | 0.028          | 0.000    | 0.720      |
| random direction       | –            | –              | –        | 0.721      |

TABLE 3 · Language steering under matched intervention budgets on `resid_nc_hm`'s final checkpoint; medians over 24 language–magnitude cells per arm. Effect is displacement along the arm's own direction (native) and along the shared difference-of-means direction (dm frame); hit rate is the fraction of steered decodes reaching the median projection of genuine target-language rows on the shared frame; collateral is content change on unrelated dimensions, with the random-direction row as the floor. Each arm dominates its native frame; on the shared frame classification forcing reaches 0.43× the supervised effect. Every arm's collateral sits at the random floor at these budgets.

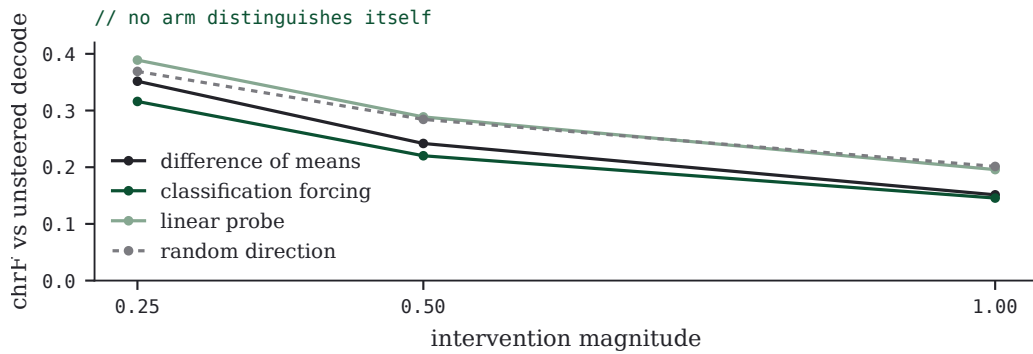


FIGURE 9 · chrF between steered and unsteered decodes by intervention magnitude. All four arms, the random direction included, damage the decode at nearly the same rate; no arm buys naturalness at matched magnitude.

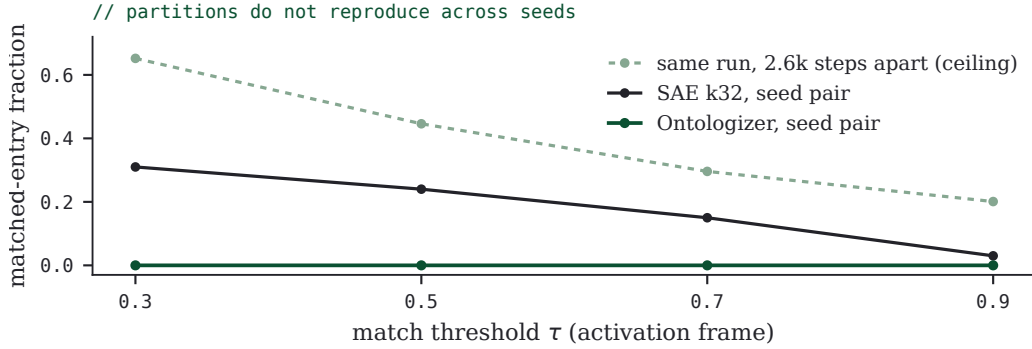


FIGURE 10 · Cross-seed matched-entry fraction on the activation frame by threshold. The same-run pair bounds what the metric can detect (0.65 at  $\tau=0.3$ ); the k32 SAE pair reproduces partially; Ontologizer partitions reproduce not at all.

architecture must beat on steering quality and task naturalness, and that comparison is where the work currently sits.

#### 6.6 SEED STABILITY

Partitions do not reproduce across seeds at matched training. A second run of the `resid_nc_hm` configuration from a different seed, compared at matched training (370k steps on both sides), matches essentially nothing (Figure 10): no head pair and at most 0.3% of entries clear a 0.3 cosine threshold on dictionary geometry, and activation-frame containment over a shared evaluation batch is at most 0.001 at any threshold, against 0.31/0.24/0.15/0.03 for the k32 SAE and its own seed replica across the four match thresholds (0.3–0.9). The metric itself has headroom: the same run compared against itself 2,600 steps apart scores 1.000 on geometry at every threshold, and 0.65 at the loosest activation threshold (falling to 0.20 at the strictest), so the activation frame has a self-ceiling well below one and the cross-seed numbers sit far below both ceilings. At matched training the Ontologizer’s partitions are less reproducible across seeds than the SAE latents this proposal criticizes. A replica carried to `resid_nc_hm`’s full 4.4M steps is the remaining control.

#### 6.7 DEVELOPMENT OVER TRAINING

Tracking `resid_nc_hm`’s full 448-checkpoint history gives the devinterp question a first answer. The tracked statistic applies Table 2’s  $z$ -score machinery to the refit end-to-end direction matrix rather than to raw dictionary decode directions, so the two are not directly comparable: the notes’ raw decode directions repel ( $z=-17.6$ , measured on `resid_nc` at 372.6k steps) while the refit directions on `resid_nc_hm`’s final checkpoint cohere ( $z=+72$ ); the statistics differ in both instrument and substrate, and the refit frame folds in the shared decoder and cross-layer interactions. On that statistic the shape is rise, peak, and partial relaxation (Figure 11): mean within-head  $z_{\cos}$  climbs from 1.8 at step 10k to a peak of 119 near step 370k (median 123, every head above  $z=2$ ), then relaxes over the remaining 4M steps to 72 at the final checkpoint (median 105, 0.79 above  $z=2$ ). The code itself moves the other way after the annealing transient: per-sample entropy returns to 4.5 bits of a 5-bit

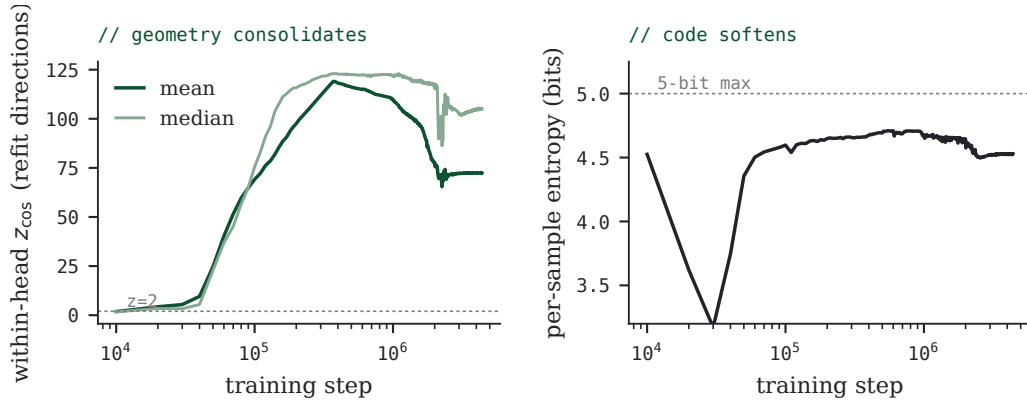


FIGURE 11 · The 448-checkpoint training history. Left: within-head coherence of refit end-to-end directions, z-scored against size-matched nulls; rise, peak near step 370k, partial relaxation. Right: mean per-sample classification entropy returns toward the 5-bit maximum after the annealing transient. Geometry consolidates while the code itself stays soft.

maximum and usage KL falls toward uniform: relative to initialization the consolidation is in the geometry the classifications write, while the classifications themselves stay soft.

#### 6.8 HSIC-BOTTLENECK ABLATION

An HSIC bottleneck is a candidate replacement for the auxiliary losses: maximize dependence with the reconstruction target while penalizing dependence between heads. Both arms are in, retrained on the `resid_nc_hm` configuration to 372k steps. Replacing the four auxiliary terms with the HSIC objective destroys head health: a late, persistent freeze takes 77 of 160 heads by the final checkpoint, one layer entirely (layer 3, 32/32 from step 80k), while reconstruction lands slightly behind the control (tail-mean training MSE  $3.7 \times 10^{-5}$  against  $3.4 \times 10^{-5}$ ). Adding HSIC on top of the auxiliary terms is worse on both counts: reconstruction  $4.4 \times 10^{-5}$ , and 41 of 160 heads still frozen at the end, layer 3 fully frozen by 80k with only partial thaw (Figure 12). The failure mode is `resid_nc`'s late freeze, not the transient, which places the bottleneck and the mean-entropy bonus in direct opposition: the bonus prevents exactly what HSIC induces. As formulated the bottleneck is not a replacement, and it is not an addition either.

#### 6.9 AGAINST THE DESIDERATA

Scored against the introduction: causal is currently positive (the forced-classification steering result is causally verified), composable is positive (interventions compose approximately additively), and closed-form is negative (parameter-space decodes fail as descriptions). Seed stability is measured on both sides: the k32 SAE pair matches at 0.31 down to 0.03 as the threshold tightens while Ontologizer partitions match at zero (Section 6.6). The residual stack's one quantified credit remains roughly  $3\times$  adaptive compensation under truncation (the shrinkage decomposition in the working notes). Of the outstanding comparisons, the steering overlay is a qualified positive, task naturalness is so far a wash, and seed stability is a clear negative; the reconstruction price Table 1 puts on the architecture stands.

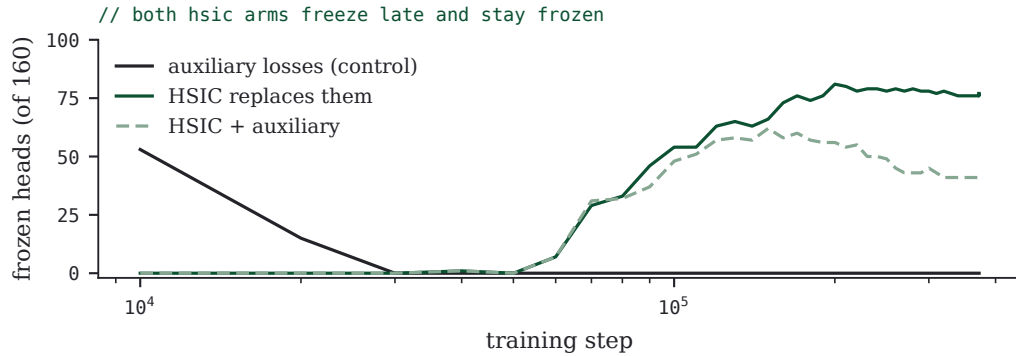


FIGURE 12 · Total frozen heads over training for the three loss configurations, all on the `resid_nc_hm` configuration at 372k steps. The auxiliary-loss control’s freeze is the early transient and resolves; both HSIC arms develop the late, persistent freeze instead, and the auxiliary terms only partially thaw it.

## 7 FUTURE WORK

### 7.1 COMPLETING THE EVALUATION

The steering overlay, task naturalness, partition-level seed stability, and the training-history diagnostics are reported above; cross-model feature splitting and sparse language probing are covered in the notes and not repeated here. What remains of the deciding set: human scoring of the blinded naturalness sheet, the full-length seed replica, and the fresh-encoded evaluation set. The last is blocked: the deterministic `mC4` stream is exhausted by the 4M-sample training cache, so no held-out tail exists and closing the train/eval asymmetry requires a different source (the `C4` validation split is the leading candidate). The controlled retrains and the HSIC arms are reported above (Figures 8 and 12). Beyond the deciding set, three comparisons remain open: the grouped-softmax rival on steering quality, the RL decode-cycle tier and its convergence race, and full-length scaling retrains.

### 7.2 CHANGE IN INTERPRETABILITY OVER TRAINING

First numbers are in Section 6.7. The remaining work is aligning the per-layer sharpening curves with independently estimated stage boundaries, which needs the boundary estimator rather than new tracking machinery.

### 7.3 TRANSCODER ADAPTATION

The architecture currently reconstructs an embedding. Adapting it to map between layers, as transcoders do, is the obvious next target.

### 7.4 HSIC BOTTLENECK

The measured ablation is in Section 6.8. Whether a reformulated bottleneck can coexist with the load balancing the mean-entropy bonus provides is open.

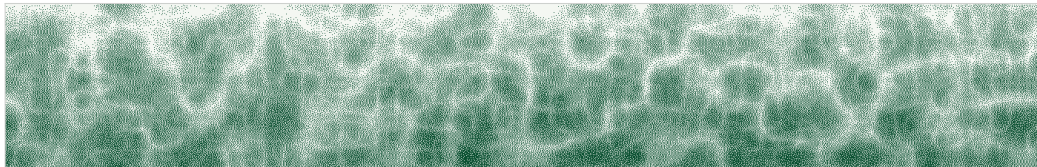
## ACKNOWLEDGEMENTS

Design, typesetting, and editorial production of this preprint are by The Flock Capital Research Department, whose error-diffusion pipeline also printed the plates.

## REFERENCES

- [1] Trenton Bricken et al. “Towards Monosemanticity: Decomposing Language Models With Dictionary Learning”. In: *Transformer Circuits Thread* (2023). URL: <https://transformer-circuits.pub/2023/monosemantic-features/index.html>.
- [2] Bart Bussmann et al. *Learning Multi-Level Features with Matryoshka Sparse Autoencoders*. 2025. arXiv: 2503.17547 [cs.LG]. URL: <https://arxiv.org/abs/2503.17547>.
- [3] Tue M. Cao et al. *Tree SAE: Learning Hierarchical Feature Structures in Sparse Autoencoders*. 2026. arXiv: 2605.07922 [cs.LG]. URL: <https://arxiv.org/abs/2605.07922>.
- [4] David Chanin et al. *A is for Absorption: Studying Feature Splitting and Absorption in Sparse Autoencoders*. 2025. arXiv: 2409.14507 [cs.CL]. URL: <https://arxiv.org/abs/2409.14507>.
- [5] Bilal Chughtai, Lawrence Chan, and Neel Nanda. *A Toy Model of Universality: Reverse Engineering How Networks Learn Group Operations*. 2023. arXiv: 2302.03025 [cs.LG]. URL: <https://arxiv.org/abs/2302.03025>.
- [6] Evan Duan. *Pre-Intervention Prediction of Sparse Autoencoder Steering Side Effects*. 2026. arXiv: 2606.08365 [cs.LG]. URL: <https://arxiv.org/abs/2606.08365>.
- [7] Paul-Ambroise Duquenne, Holger Schwenk, and Benoît Sagot. *SONAR: Sentence-Level Multimodal and Language-Agnostic Representations*. 2023. arXiv: 2308.11466 [cs.CL]. URL: <https://arxiv.org/abs/2308.11466>.
- [8] Leo Gao et al. *Scaling and evaluating sparse autoencoders*. 2024. arXiv: 2406.04093 [cs.LG]. URL: <https://arxiv.org/abs/2406.04093>.
- [9] Jesse Hoogland et al. *The Developmental Landscape of In-Context Learning*. 2024. arXiv: 2402.02364 [cs.LG]. URL: <https://arxiv.org/abs/2402.02364>.
- [10] Adam Karvonen et al. *SAEBench: A Comprehensive Benchmark for Sparse Autoencoders in Language Model Interpretability*. 2025. arXiv: 2503.09532 [cs.LG]. URL: <https://arxiv.org/abs/2503.09532>.
- [11] Anton Korznikov et al. *OrtSAE: Orthogonal Sparse Autoencoders Uncover Atomic Features*. 2025. arXiv: 2509.22033 [cs.LG]. URL: <https://arxiv.org/abs/2509.22033>.
- [12] Leon Lang, Lawrence Chan, and Erik Jenner. *Natural Abstractions: Key Claims, Theorems, and Critiques*. 2023. URL: <https://www.lesswrong.com/posts/gvzW46Z3BsaZsLc25/natural-abstractions-key-claims-theorems-and-critiques-1>.
- [13] Aleksandar Makelov, Georg Lange, and Neel Nanda. *Is This the Subspace You Are Looking for? An Interpretability Illusion for Subspace Activation Patching*. 2023. arXiv: 2311.17030 [cs.LG]. URL: <https://arxiv.org/abs/2311.17030>.
- [14] Anish Mudide et al. *Efficient Dictionary Learning with Switch Sparse Autoencoders*. 2025. arXiv: 2410.08201 [cs.LG]. URL: <https://arxiv.org/abs/2410.08201>.
- [15] Neel Nanda et al. *A Pragmatic Vision for Interpretability*. 2025. URL: <https://www.alignmentforum.org/posts/StENzDcD3kpFGJssR/a-pragmatic-vision-for-interpretability>.
- [16] Neel Nanda et al. *Progress measures for grokking via mechanistic interpretability*. 2023. arXiv: 2301.05217 [cs.LG]. URL: <https://arxiv.org/abs/2301.05217>.
- [17] Kiho Park et al. *The Geometry of Categorical and Hierarchical Concepts in Large Language Models*. 2025. arXiv: 2406.01506 [cs.CL]. URL: <https://arxiv.org/abs/2406.01506>.

- [18] Gonalo Paulo et al. *Automatically Interpreting Millions of Features in Large Language Models*. 2025. arXiv: 2410.13928 [cs.LG]. URL: <https://arxiv.org/abs/2410.13928>.
- [19] Michael T. Pearce et al. *Bilinear MLPs enable weight-based mechanistic interpretability*. 2025. arXiv: 2410.08417 [cs.LG]. URL: <https://arxiv.org/abs/2410.08417>.
- [20] LCM team et al. *Large Concept Models: Language Modeling in a Sentence Representation Space*. 2024. arXiv: 2412.08821 [cs.CL]. URL: <https://arxiv.org/abs/2412.08821>.
- [21] Andrew Wagenmaker et al. *Steering Your Diffusion Policy with Latent Space Reinforcement Learning*. 2025. arXiv: 2506.15799 [cs.R0]. URL: <https://arxiv.org/abs/2506.15799>.
- [22] Zhengxuan Wu et al. *AxBench: Steering LLMs? Even Simple Baselines Outperform Sparse Autoencoders*. 2025. arXiv: 2501.17148 [cs.CL]. URL: <https://arxiv.org/abs/2501.17148>.
- [23] Zhengyan Zhang et al. *MoEfication: Transformer Feed-forward Layers are Mixtures of Experts*. 2022. arXiv: 2110.01786 [cs.CL]. URL: <https://arxiv.org/abs/2110.01786>.



RIDGE · SEED 32 · THE SAME PIPELINE, CLOSING THE FILE